



Numerical investigation on the molten pool and keyhole dynamic behaviors and weld microstructure in laser-induction hybrid welding of stainless steel

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ABSTRACT

The laser-induction hybrid welding (LIHW) can inhibit the formation of welding defects and enhance the mechanical properties of welded joints, which improves the applicability of laser welding in different industries. In this paper, a macro-micro numerical model consisting of macroscopic heat transfer and fluid flow model coupled with magnetic field, transient solidification conditions (SCs) model and microscopic phase field model (PFM) is developed to investigate the LIHW of 304 stainless steel. The validity of the developed model is confirmed by comparing the simulation results with the experimental results. The effects of electromagnetic induction heating on the molten pool and keyhole dynamic behaviors and weld microstructure during LIHW are analyzed and discussed in detail. Compared with the single-laser welding (SLW), the depth and half width of the molten pool are increased and the stability of the keyhole has been improved during LIHW. Additionally, the primary dendrite arm spacing during SLW is smaller than that during LIHW. The results show that the proposed model is beneficial for understanding the molten pool and keyhole dynamic behaviors and microstructure evolution process during LIHW and hence improving the welding quality.

1. Introduction

Stainless steel materials are utilized in nuclear energy, aerospace and other industries extensively due to the outstanding mechanical properties, high specific strength and corrosion resistance [1–3]. Laser welding is an effective way of joining stainless steel materials owing to the narrow heat zone and high energy density [4,5]. However, the existing welding defects such as pores and cracks are often formed during the laser welding process [6,7]. Recently, the laser-induction hybrid welding (LIHW) method is proposed and has been indicated to be able to improve the joint ductility and inhibit the welding defects [8]. Compared to the conventional heat treatments, the electromagnetic induction heating has the advantages of high energy utilization, high heating speed and high safety [9]. As an auxiliary heat source, electromagnetic induction heating also can extend the solidification time of molten pool and has great effect on the weld microstructure [10]. Therefore, it is of importance to analyze the molten pool and keyhole dynamic behaviors and microstructure evolution process during LIHW for improving the microstructure and joint quality.

Lots of scholars have studied the molten pool and keyhole dynamic behaviors and the formation of welding defects during the laser welding

process. Li et al. [11] established the mathematical model to calculate the keyhole dynamic behavior and the formation of pores during laser welding of high strength low alloy steel. Their results showed that a few bubbles formed near the fusion line were likely to be absorbed by the solidification surface, which resulted in some porosities appearing near the fusion line. Pang et al. [12] investigated the coupling of heat transfer, fluid flow and keyhole free surface variations during laser welding of aluminum alloy. The obtained results revealed that the stability of keyhole was significantly affected by the molten pool dynamic behaviors. Zhang et al. [13] developed the three-dimensional laser deep penetration welding model coupled with the volume of fluid method and ray-tracing algorithm during laser welding of low carbon steel. They discussed the periodic characteristics of the dynamic behaviors of molten pool and keyhole during the laser welding process. Lin et al. [14] analyzed the keyhole dynamic behavior in the remote laser welding process of aluminum alloy. Their obtained results showed that the dynamic behaviors of molten pool and keyhole were closely related to the formation of pores. They found that the formation of pores was mainly due to the high flow velocity of molten metal behind the keyhole.

Due to the significant influence of weld microstructure on welding quality, many experimental and numerical investigations have been

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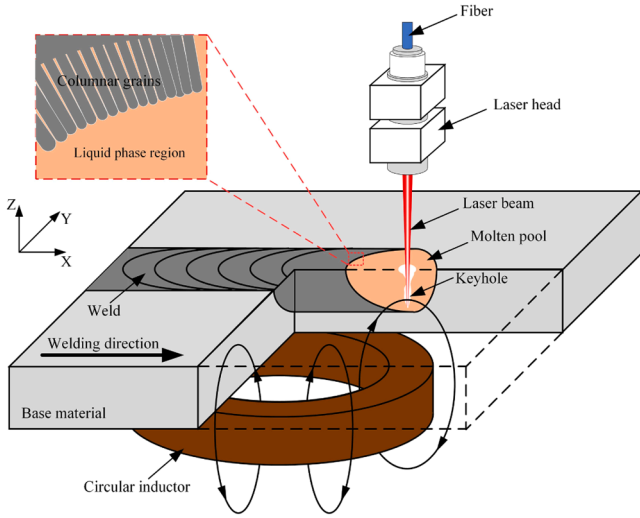


Fig. 1. The schematic diagram of the laser welding process.

conducted to explore the microstructure evolution process during laser welding. Yao et al. [15] analyzed the microstructure near the interface between copper sheet and intermixing region in laser butt welding of dissimilar materials of steel and copper by optical microscopy and scanning electron microscopy. Their results showed that transition regions with numerous gradual phases appeared in the upper part of interface. Liu et al. [16] investigated the influence of welding process parameters on the weld microstructure during the high-power laser welding process of stainless steel. They pointed out that the microstructure characteristics were significantly influenced by temperature gradient (TG) and solidification rate (SR). Zheng et al. [17] developed a Monte-Carlo-assisted phase field model (PFM) for calculating the grain growth during laser welding of dissimilar materials of Fe-Si alloy and low carbon steel. The results demonstrated that the columnar grains were observed in the heat-affected zone and the corresponding average grain size was larger than that in the fusion zone. Ai et al. [18] analyzed the microstructure evolution process during the oscillating laser welding of aluminum alloy by the PFM with the dynamic TG and SR. They investigated the effect of flow field on the microstructure evolution process in oscillating laser welding and found that the fluctuation of TG played a major role in the transformation of grain morphology.

The principle of electromagnetic induction heating is that the eddy currents induced inside the metal under a high-frequency alternating magnetic field generate the Joule heat for metal absorption [19]. The electromagnetic induction heating technology has been introduced into the welding industry recently. Gao et al. [20] developed a finite element model for analyzing the effect of inductor position on temperature uniformity during the induction lap welding process of dissimilar materials of aluminum alloy and steel. The results indicated that the maximum temperature of base material increased and the temperature uniformity decreased under the condition of small distance between the inductor and the center of the lap edge. Li et al. [21] explored the improvement mechanism of LIHW on weld formation by comparing the vapor/plasma plume and keyhole dynamic behaviors between single-laser welding (SLW) and LIHW process of high strength low alloy steel. They found that the spatter defect was inhibited, the width and length of molten pool were increased during LIHW with the increase of inductor output power. They [22] also investigated the mechanical properties and microstructure of high strength low alloy steel welded joints through the mechanical testing, optical microscopy, electron backscatter diffraction and scanning electron microscopy. Their results demonstrated that the formation of welding defects such as pores and cracks was inhibited during LIHW. They found that the grain average size increased and the microhardness decreased under the preheating

effect of induction heating. Coelho et al. [23] studied the influence of multiple induction-assisted strategies on the microstructure during LIHW of high strength low alloy steel. Their results showed that the cooling time was increased significantly, the formation of martensite within the fusion zone and heat-affected zone was inhibited during LIHW compared to that during SLW. Pinto et al. [24] analyzed the residual stresses and microstructure of high strength low alloy steel welded joints during LIHW. The obtained results indicated that the formation of martensite was inhibited and the hardness within fusion zone was decreased effectively under the effect of induction heating.

From the previous researches, LIHW has been proven to be an effective laser hybrid welding method to inhibit welding defects and improve welding quality. However, the effects of electromagnetic induction heating on the molten pool and keyhole dynamic behaviors and weld microstructure during LIHW have not been studied well. Therefore, a macro-micro numerical model consisting of macroscopic heat transfer and fluid flow model coupled with magnetic field, transient solidification conditions (SCs) model and microscopic PFM during LIHW of 304 stainless steel is developed to calculate the molten pool and keyhole dynamic behaviors and microstructure evolution process. The molten pool and keyhole dynamic behaviors and microstructure evolution process during LIHW and SLW have been compared in this paper. The effects of electromagnetic induction heating on the molten pool and keyhole dynamic behaviors and weld microstructure are analyzed in detail. The results can provide the guidance for revealing the weld formation and microstructure evolution mechanism during LIHW and hence improving the welded joints performance.

2. Numerical modelling

2.1. Macroscopic heat transfer and fluid flow model coupled with magnetic field

The schematic diagram of the laser welding process is shown in Fig. 1. The macroscopic heat transfer and fluid flow model coupled with magnetic field is established to obtain the temperature field, flow field and magnetic field. In the model, the Lorentz force acting on the molten metal is not taken into account. The melting and solidification processes are described by melting-solidification model involving the enthalpy porosity method which is used to treat the mushy zone in the molten pool as porous media [25]. The main governing equations including the continuity equation, momentum equation and energy equation are referred from the previous literature [18].

The volume of fluid method is used to track the gas-liquid interface during laser welding. The surface tension, Marangoni force and recoil pressure are considered in the model.

Volume of fluid equation [26]:

$$\frac{\partial F}{\partial t} + \nabla \cdot (\vec{V}F) = 0 \quad (1)$$

where F is the volume of fluid function. $\vec{V}$ is the velocity vector.

Surface tension P_s [27,28]:

$$P_s = \kappa \gamma \quad (2)$$

$$\gamma = \gamma_0 + \frac{d\gamma}{dT}(T - T_l) \quad (3)$$

where κ , γ , γ_0 and $d\gamma/dT$ are the free surface curvature, surface tension coefficient, surface tension coefficient of pure metal at the liquidus temperature T_l and temperature coefficient of surface tension, respectively.

Marangoni force F_M [29]:

$$F_M = \frac{\partial T}{\partial s} \frac{\partial \gamma}{\partial T} \quad (4)$$

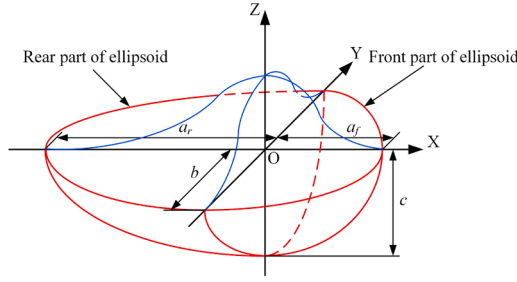


Fig. 2. The schematic diagram of the double ellipsoid heat source model.

where $\vec{s}$ is the tangent vector.

Recoil pressure F_p [30]:

$$F_p = \frac{A_p B_0}{\sqrt{T}} \exp\left(-\frac{M_a L_v}{N_a k_b T}\right) \quad (5)$$

where A_p and B_0 are the numerical coefficient and evaporation constant, respectively. M_a is the atomic mass and L_v is the latent heat of evaporation. N_a is the Avogadro's number. k_b is the Boltzmann's constant.

The energy loss caused by the convection and radiation can be calculated as [28]:

$$k_r \frac{\partial T}{\partial n} = -h_c(T - T_r) - \varepsilon_0 \sigma_0(T^4 - T_r^4) \quad (6)$$

where k_r , h_c , T_r , ε_0 and σ_0 are the thermal conductivity, convection coefficient, reference temperature, surface radiation emissivity and Stefan-Boltzmann constant, respectively.

A ray tracing heat source model is adopted to characterize the laser beam energy absorbed by the base material. The laser beam radiated on the upper surface of the base material is discretized to describe the energy absorption on the keyhole wall accurately. During the laser beam multiple reflection process, the Fresnel absorption of laser beam energy by the base material can be expressed as [31]:

$$q_F = I_0(r, z) \left(\vec{I}_0 \cdot \vec{n}_0 \right) \alpha_{Fr}(\theta_0) + \sum_{n=1}^N I_n(r, z) \left(\vec{I}_n \cdot \vec{n}_n \right) \alpha_{Fr}(\theta_n) \quad (7)$$

where $I_0(r, z)$ is the beam energy density distribution before irradiation. $\vec{I}$ is the unit vector of incident beams. $\vec{n}$ is the unit normal vector. α_{Fr} is the Fresnel absorption coefficient. θ is the angle between the incident beam and the normal to the keyhole wall. N is the incident times of beams after multiple reflections. $I_n(r, z)$ is the beam energy density of the n -th reflections.

The inverse Bremsstrahlung absorption of laser beam energy by the

plasma induced in the laser welding process can be described as [31]:

$$I_n(r, z) = I_r(r, z) \exp\left(-\int_0^{l_n} K_{pl} dl\right) \quad (8)$$

where $I_r(r, z)$ is the beam energy density at the current intersection point between the ray and keyhole. l_n is the distance from the current intersection point to the next intersection point. K_{pl} is the inverse bremsstrahlung absorption coefficient. The distribution of laser beam energy can be expressed as [31,32]:

$$I_0(r, z) = \frac{3f_1 P_a}{\pi r_0^2} \exp\left(-\frac{3r^2}{r_0^2}\right) \quad (9)$$

$$r = \sqrt{x^2 + y^2} \quad (10)$$

where f_1 , P_a and r_0 are the energy distribution coefficient of ray tracing heat source, laser power and heat source radius, respectively.

As shown in Fig. 2, a double ellipsoid heat source model is applied to characterize the distribution of radiation energy of metal vapor and plasma on the upper surface of the molten pool [33]:

$$q_f(x, y, z) = \frac{6\sqrt{3}f_f f_2 P_a}{a_f b c \pi \sqrt{\pi}} \exp\left[-3\left(\frac{x^2}{a_f^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)\right] \quad (11)$$

$$q_r(x, y, z) = \frac{6\sqrt{3}f_r f_2 P_a}{a_r b c \pi \sqrt{\pi}} \exp\left[-3\left(\frac{x^2}{a_r^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)\right] \quad (12)$$

$$f_f + f_r = 2 \quad (13)$$

$$f_1 + f_2 = 1 \quad (14)$$

where $q_f(x, y, z)$ and $q_r(x, y, z)$ are the power densities in front and rear parts of double ellipsoid heat source model, respectively. f_f and f_r are the energy distribution coefficients in front and rear parts of double ellipsoid heat source model, respectively. f_2 is the energy distribution coefficient of double ellipsoid heat source. a_f , b , c and a_r are the shape parameters of double ellipsoid heat source model, respectively.

The electromagnetic field during LIHW is governed by Maxwell's equations [34,35]:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (15)$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad (16)$$

$$\nabla \cdot \vec{D} = \rho_e \quad (17)$$

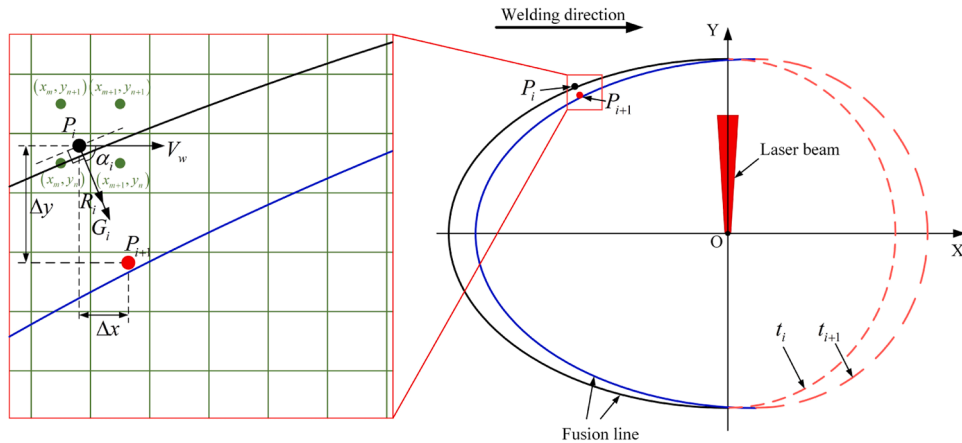


Fig. 3. The schematic diagram of the transient SCs model.

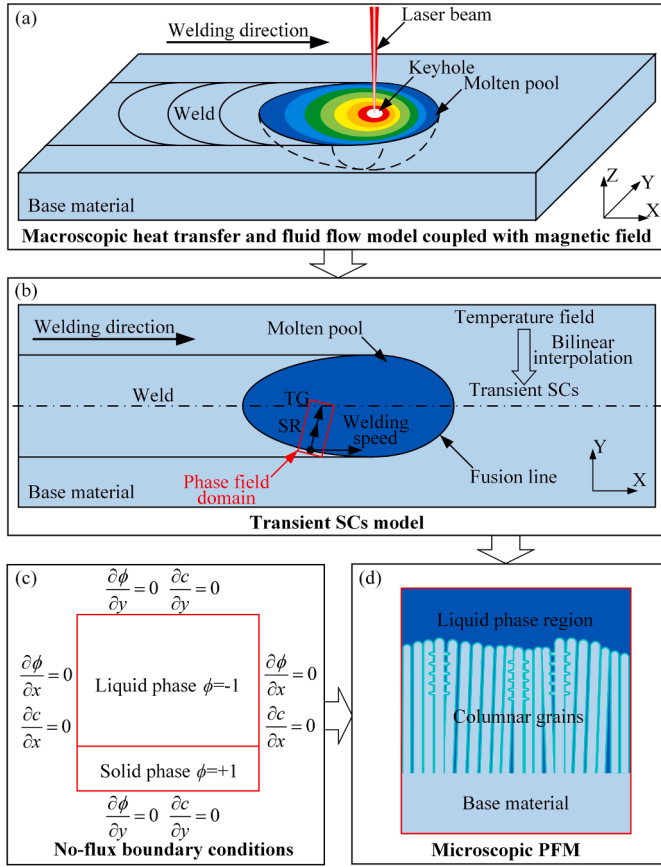


Fig. 4. The schematic diagram of the numerical implementation procedure: (a) Macroscopic heat transfer and fluid flow model coupled with magnetic field; (b) Transient SCs model; (c) No-flux boundary conditions; (d) Microscopic PFM.

$$\nabla \cdot \vec{B} = 0 \quad (18)$$

where $\vec{E}$, $\vec{B}$, $\vec{H}$, $\vec{J}$, $\vec{D}$ and ρ_e are the electric field vector, magnetic flux density vector, magnetic field intensity vector, current density vector, electric flux density vector and electric charge density, respectively.

The constitutive equations are as follows [36]:

$$\vec{D} = \epsilon \vec{E} \quad (19)$$

$$\vec{B} = \mu_r \mu_0 \vec{H} \quad (20)$$

$$\vec{J} = \sigma \vec{E} \quad (21)$$

where ϵ , μ_r , μ_0 and σ are the permittivity, relative permeability, permeability in vacuum and electrical conductivity, respectively.

The Joule heat q_e generated by induced eddy current in base material can be expressed as [37]:

$$q_e = \frac{|\vec{J}|^2}{\sigma} \quad (22)$$

2.2. Transient SCs model

Based on the calculated temperature field by the macroscopic heat transfer and fluid flow model coupled with magnetic field, the transient SCs model is developed to obtain the transient TG and SR near the fusion line on the horizontal section used for phase field simulation. The schematic diagram of the transient SCs model is shown in Fig. 3. The effect of horizontal section height on the transient SCs near the fusion

line is not considered in the transient SCs model. The transient TG G_i and SR R_i at arbitrary point P_i (dendrite tip position) near the fusion line are defined as [38]:

$$G_i = \sqrt{G_{x_i}^2 + G_{y_i}^2} \quad (23)$$

$$R_i = V_w \cos \alpha_i \quad (24)$$

$$G_{x_i} = \frac{1}{(x_{m+1} - x_m)(y_{n+1} - y_n)} [x_{m+1} - x_i \quad x_i - x_m] \times \begin{bmatrix} G_{x_{m,n}} & G_{x_{m,n+1}} \\ G_{x_{m+1,n}} & G_{x_{m+1,n+1}} \end{bmatrix} \begin{bmatrix} y_{n+1} - y_i \\ y_i - y_n \end{bmatrix} \quad (25)$$

$$G_{y_i} = \frac{1}{(x_{m+1} - x_m)(y_{n+1} - y_n)} [x_{m+1} - x_i \quad x_i - x_m] \times \begin{bmatrix} G_{y_{m,n}} & G_{y_{m,n+1}} \\ G_{y_{m+1,n}} & G_{y_{m+1,n+1}} \end{bmatrix} \begin{bmatrix} y_{n+1} - y_i \\ y_i - y_n \end{bmatrix} \quad (26)$$

$$\alpha_i = \arctan \frac{-G_{y_i}}{G_{x_i}} \quad (27)$$

where G_{x_i} and G_{y_i} are the components of G_i along X and Y directions, respectively. (x_i, y_i) is the coordinate of arbitrary point P_i near the fusion line. V_w is the welding speed. α_i is the angle between welding speed and TG. (x_m, y_n) , (x_m, y_{n+1}) , (x_{m+1}, y_n) and (x_{m+1}, y_{n+1}) are the coordinates of four grid center positions near the arbitrary point P_i on the horizontal section of the computational domain of macroscopic heat transfer and fluid flow model coupled with magnetic field. After the interval time dt , the x-coordinate x_{i+1} and y-coordinate y_{i+1} of the point P_{i+1} near the fusion line can be described as:

$$x_{i+1} = x_i + \Delta x \quad (28)$$

$$\Delta x = V_w (\cos \alpha_i)^2 dt \quad (29)$$

$$y_{i+1} = y_i - \Delta y \quad (30)$$

$$\Delta y = V_w \cos \alpha_i \sin \alpha_i dt \quad (31)$$

2.3. Microscopic PFM

For investigating the microstructure evolution process, a microscopic PFM is developed based on the model proposed by Echebarria et al. [39]. The corresponding governing equations can be described as:

$$\left[1 - (1-k) \frac{Y_s - \int_0^t \tilde{R}(t') dt'}{\tilde{l}_r} \right] a(\mathbf{n}) \frac{\partial \phi}{\partial t} = \nabla \cdot [a(\mathbf{n})^2 \nabla \phi] + \partial_x \left[|\nabla \phi|^2 a(\mathbf{n}) \frac{\partial a(\mathbf{n})}{\partial (\partial_x \phi)} \right] + \partial_y \left[|\nabla \phi|^2 a(\mathbf{n}) \frac{\partial a(\mathbf{n})}{\partial (\partial_y \phi)} \right] + \phi - \phi^3 - \lambda (1 - \phi^2)^2 \left[U + \frac{Y_s - \int_0^t \tilde{R}(t') dt'}{\tilde{l}_r} \right] \quad (32)$$

$$\left(\frac{1+k}{2} - \frac{1-k}{2} \phi \right) \frac{\partial U}{\partial t} = \nabla \cdot \left\{ \tilde{D}_r \frac{1-\phi}{2} \nabla U + \frac{1}{2\sqrt{2}} [1 + (1-k)U] \frac{\partial \phi}{\partial t} \frac{\nabla \phi}{|\nabla \phi|} \right\} + \frac{1}{2} \frac{\partial \phi}{\partial t} [1 + (1-k)U] \quad (33)$$

with

$$U = \frac{1}{1-k} \left\{ \frac{2kc}{c_\infty [1 - \phi + k(1 + \phi)]} - 1 \right\} \quad (34)$$

where k is the partition coefficient. Y_s is the axis along the direction of columnar grain growth. $a(\mathbf{n})$ is the anisotropy of surface tension. ϕ is the

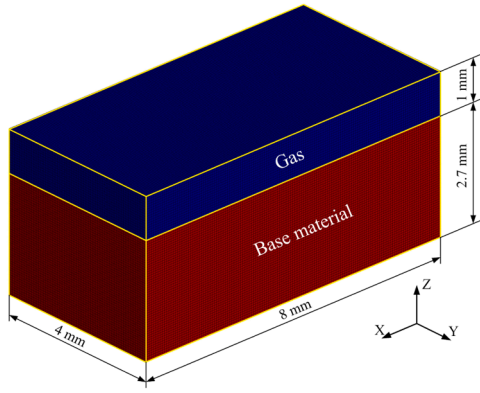


Fig. 5. The mesh model of computational domain.

Table 1

The physical parameters used in the numerical calculation [40–45].

Parameter	Symbol	Value	Unit
Density	ρ	7200	kg/m ³
Specific heat	C_p	712	J/(kg K)
Thermal conductivity	k_r	35	W/(m K)
Latent heat of fusion	L_f	6.0×10^5	J/kg
Latent heat of evaporation	L_v	6.52×10^6	J/kg
Solidus temperature	T_s	1697	K
Liquidus temperature	T_l	1727	K
Evaporation temperature	T_e	3100	K
Viscosity	μ_v	0.006	kg/(m s)
Temperature coefficient of surface tension	$d\gamma/dT$	-4.3×10^{-4}	N/(m K)
Thermal expansion coefficient	β	4.48×10^{-5}	K ⁻¹
Relative permeability	μ_r	1.3	/
Electrical conductivity	σ	7.7×10^5	S/m
Partition coefficient	k	0.83	/
Initial solute concentration	c_∞	19	wt. %
Liquidus slope	m	-4.9	K/(wt. %)
Gibbs-Thomson coefficient	Γ	3.2×10^{-7}	m K

phase field parameter. $\phi = +1$ corresponds to the solid phase and $\phi = -1$ corresponds to the liquid phase. λ is the coupling factor defined as $\lambda = a_1 W/d_0$ where $a_1 = 5\sqrt{2}/8$. W and d_0 are the diffusion interface width and capillarity length. d_0 can be expressed as $d_0 = \Gamma/[|m|c_\infty(1/k - 1)]$. Γ , m and c_∞ are the Gibbs-Thomson coefficient, liquidus slope and initial solute concentration, respectively. c is the solute concentration. The dimensionless SR $\tilde{R}$, dimensionless thermal length $\tilde{l}_T$ and dimensionless liquid diffusion coefficient $\tilde{D}_l$ can be described as:

$$\tilde{R} = \frac{R\tau_0}{W} \quad (35)$$

$$\tilde{l}_T = \frac{|m|c_\infty(1/k - 1)}{WG} \quad (36)$$

$$\tilde{D}_l = \frac{D_l\tau_0}{W^2} \quad (37)$$

where R , τ_0 , G and D_l are the SR, relaxation time, TG and liquid diffusion coefficient, respectively.

2.4. Numerical implementation

The schematic diagram of the numerical implementation procedure is shown in Fig. 4. From Fig. 4(a), the temperature field, flow field during SLW and temperature field, flow field, magnetic field during LIHW are calculated by using the finite volume method to solve the macroscopic heat transfer and fluid flow model coupled with magnetic field and the solving process is realized by the commercial computational fluid dynamics software Fluent and commercial finite element

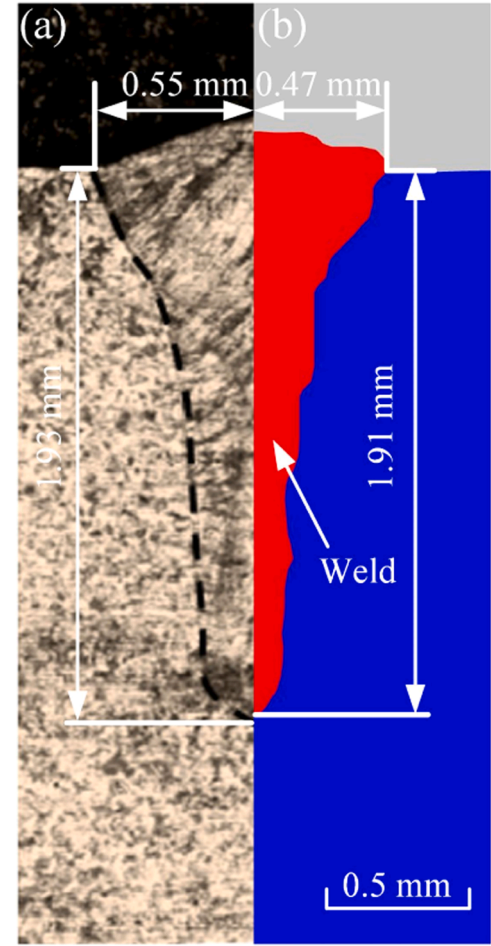


Fig. 6. The comparison of the weld cross sections obtained from the experiment and simulation: (a) Experiment result [40]; (b) Simulation result.

software COMSOL. The welding process parameters selected in this paper are laser power 1000 W, welding speed 3 m/min and defocus distance 0 mm. During the welding process, the circular inductor moves synchronously with the laser head. The intensity and frequency of input current in the circular inductor are 3500 A and 100 kHz, respectively. The inner diameter, outer diameter and thickness of circular inductor are 60 mm, 80 mm and 10 mm, respectively. The gap between circular inductor and base material is set as 5 mm. The number of turns of coil in the circular inductor is 4. The mesh model of computational domain is shown in Fig. 5. From Fig. 5, the dimension of computational domain is 8 mm $\times$ 4 mm $\times$ 3.7 mm. The upper part of the computational domain is set as the gas layer with a thickness of 1 mm. The lower part of the computational domain is the base material layer with a thickness of 2.7 mm. To solve the transient SCs model during the calculation process, the user-defined functions are adopted to extract the transient TG and SR near the fusion line, as shown in Fig. 4(b). From Fig. 4(c), no-flux boundary conditions are adopted for the microscopic PFM. The microscopic PFM is solved by the finite difference method to obtain the microstructure evolution process and the solving process is realized by the Python scripts, as shown in Fig. 4(d). The physical parameters of 304 stainless steel used in the numerical calculation are shown in Table 1.

3. Results and discussion

3.1. Model validation

The weld cross sections obtained from the experiment and simulation are compared under the same welding process parameters of laser power

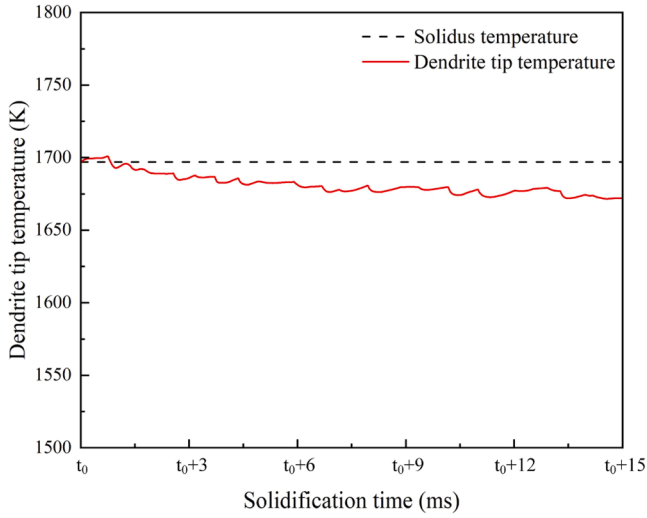


Fig. 7. The temperature variation curve of dendrite tip with solidification time.

1000 W, welding speed 3 m/min and defocus distance 0 mm during SLW to validate the effectiveness of the macroscopic heat transfer and fluid flow model coupled with magnetic field, as shown in Fig. 6. From Fig. 6, it is clearly seen that the calculated morphology of weld cross section

from simulation agrees well with that from experiment and the relative errors of depth and half width of weld are 1.0 % and 14.5 %, respectively. The reason for errors may be that partial physical parameters used in the numerical simulation are set as the constant values which ignore the effect of the temperature variation on the calculation process. The comparing result demonstrates that the macroscopic heat transfer and fluid flow model coupled with magnetic field is valid.

The temperature variation curve of dendrite tip with solidification time at point P_i near the fusion line during SLW obtained from the transient SCs model is extracted, as shown in Fig. 7. The dendrite tip temperature decreases slightly with fluctuation as the solidification time increases in Fig. 7. The temperature value of dendrite tip always fluctuates around the solidus temperature and the fluctuation amplitude is less than 25 K, which confirms the dendrite tip position is always near the fusion line and hence the transient SCs model is effective.

The weld microstructure morphologies obtained from the experiment and simulation during SLW are also compared to verify the effectiveness of the microscopic PFM, as shown in Fig. 8. From Fig. 8, both of the weld microstructure morphologies obtained from the experiment and simulation agree well. The primary dendrite arm spacings (λ_s) from experiment and simulation are about 1.98 μm and 1.89 μm , respectively. The relative error of primary dendrite arm spacings is small. The difference between experiment and simulation may be due to the fact that the influence of grain orientation is neglected in the numerical simulation process [46]. Therefore, the effectiveness of developed model is confirmed and this model can be used for calculating the

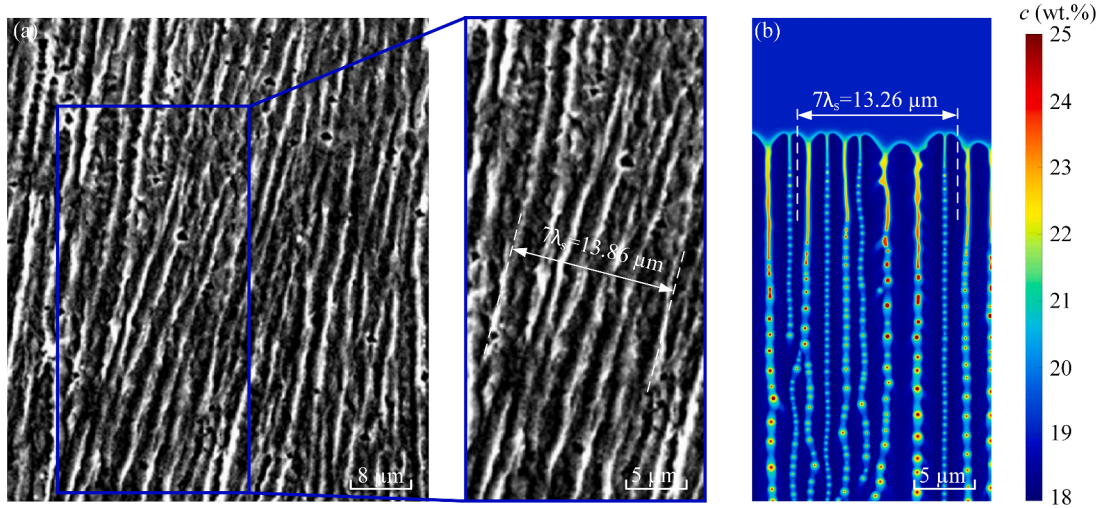


Fig. 8. The comparison of the weld microstructure morphologies obtained from the experiment and simulation: (a) Experiment result [40]; (b) Simulation result.

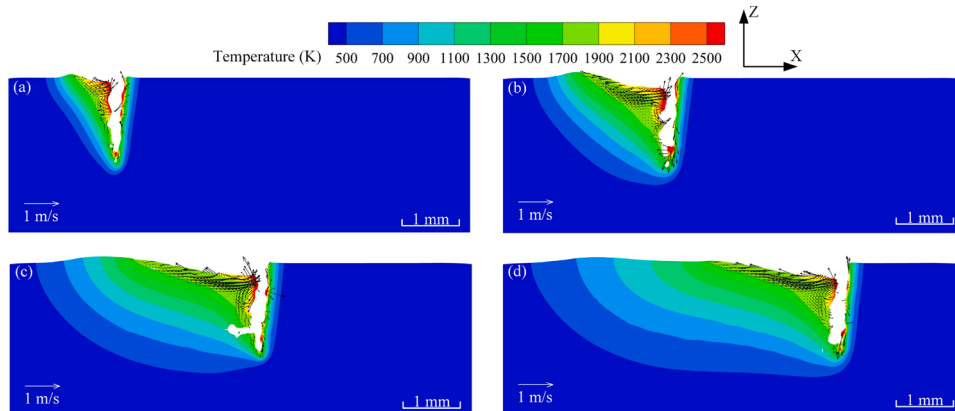


Fig. 9. The dynamic behaviors of molten pool and keyhole during SLW: (a) t_1 ; (b) t_1+20 ms; (c) t_1+50 ms; (d) t_1+80 ms.

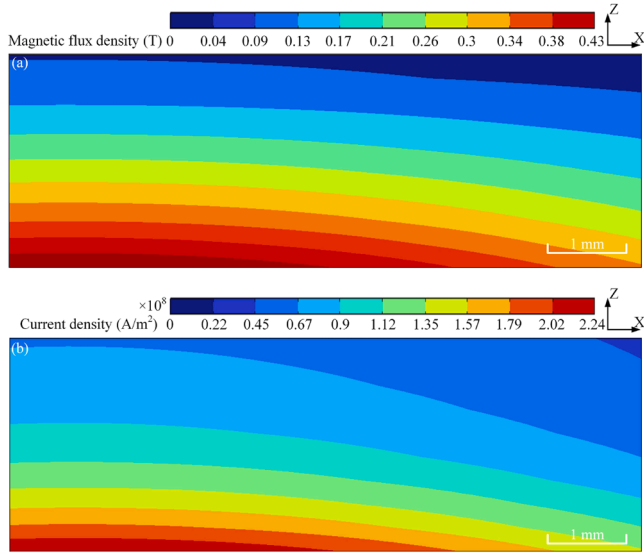


Fig. 10. The magnetic flux density distribution and current density distribution during LIHW: (a) Magnetic flux density distribution; (b) Current density distribution.

microstructure evolution process.

3.2. Dynamic behaviors of molten pool and keyhole

As the laser beam radiates on the upper surface of the base material, the temperature of base material increases gradually after absorbing the energy of laser beam. The base material melts first and then evaporates as the temperature gradually increases above the evaporation temperature. The recoil pressure of the metal vapor produced by the base material evaporation is large, which results in the keyhole formation in the molten pool. The dynamic behaviors of molten pool and keyhole during SLW are shown in Fig. 9. At t_1 , the time of laser beam radiation to the upper surface of base material is short and the absorbed laser beam energy by the molten pool is low, which results in the molten pool with short length. The flow velocity of molten metal in molten pool is small, which shows the keyhole with excellent stability, as shown in Fig. 9(a). At t_1+20 ms, the laser beam energy absorbed by the molten pool increases with the molten pool continuing to move along the welding direction, the length and depth of molten pool are increased. The flow velocity of molten metal in molten pool indicates the large increasing amplitude, especially the flow velocity of molten metal in the region with humps on the keyhole wall, as shown in Fig. 9(b). As time goes, the length of rear part of molten pool is continually increasing. The collapse is generated on the rear wall of keyhole under the effect of the molten metal flowing, which may cause the formation of pores in weld, as shown in Fig. 9(c). At t_1+80 ms, the shapes of the molten pool and keyhole are basically kept in the stable status. The rear wall of keyhole

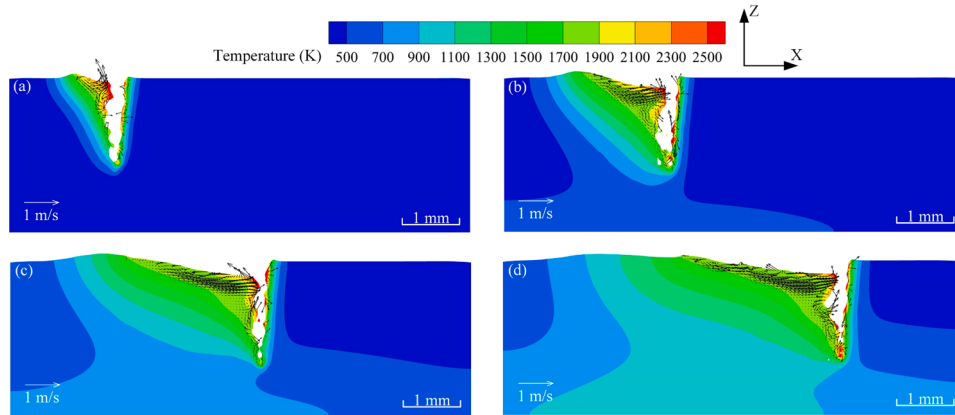


Fig. 11. The dynamic behaviors of molten pool and keyhole during LIHW: (a) t_1 ; (b) t_1+20 ms; (c) t_1+50 ms; (d) t_1+80 ms.

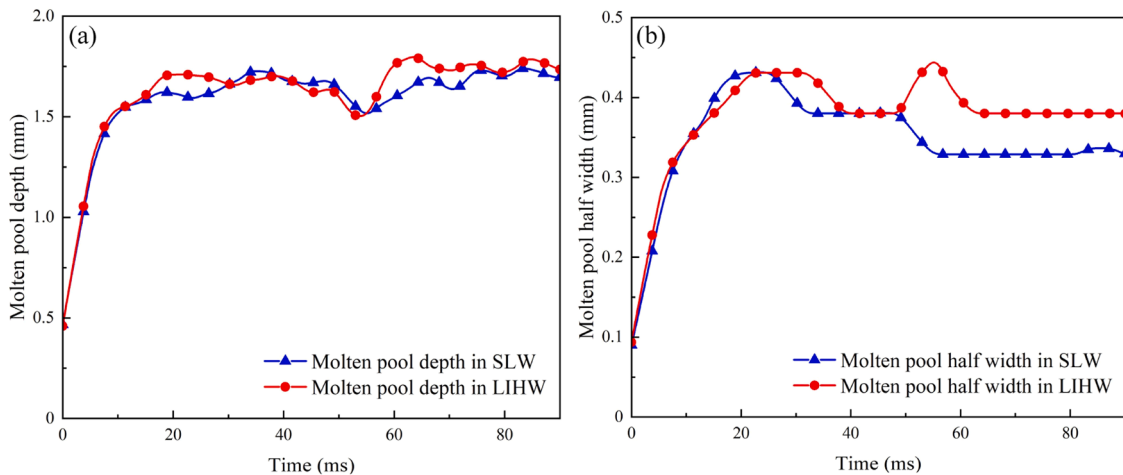


Fig. 12. The variation curves of molten pool depth and half width with time during SLW and LIHW: (a) Molten pool depth; (b) Molten pool half width.

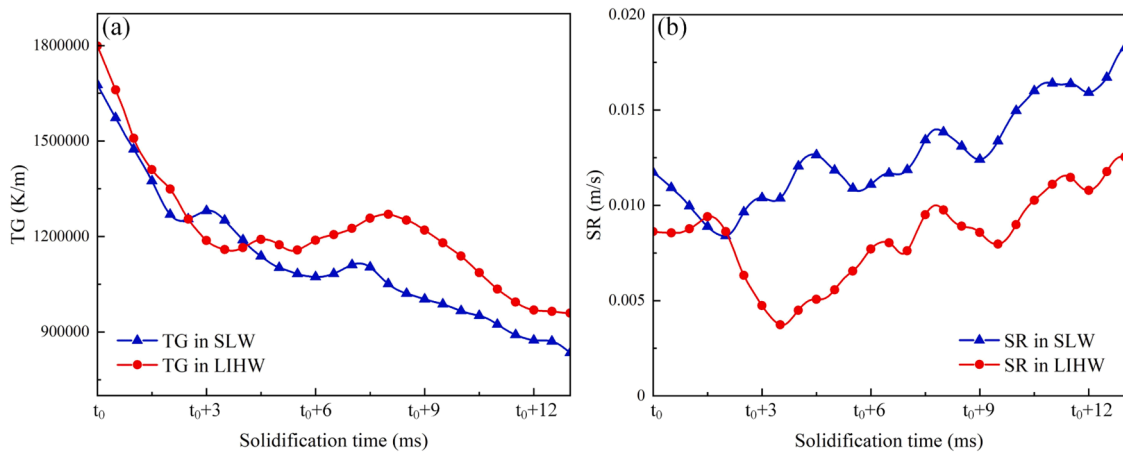


Fig. 13. The variation curves of TG and SR with solidification time during SLW and LIHW: (a) TG; (b) SR.

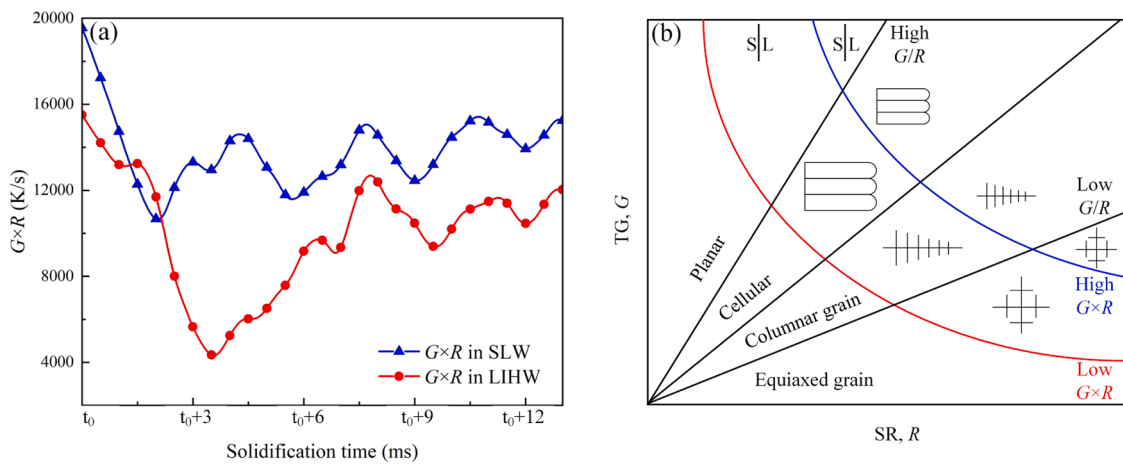


Fig. 14. The variation curves of $G \times R$ with solidification time during SLW and LIHW and the effects of SCs on the grain morphology and size: (a) The variation curves of $G \times R$; (b) The effects of SCs on the grain morphology and size [48].

gradually becomes smooth and the humping phenomenon is disappearing on the wall of keyhole, as shown in Fig. 9(d). During the SLW process, the keyhole is moving forward rapidly with fluctuation. The instable phenomena like the keyhole collapse potentially lead to the formation of pores in weld. The mechanical properties of the welded joint may be affected by the welding defects seriously.

According to the Lenz law and Faraday law, an alternating magnetic field is formed under the effect of alternating current in the circular inductor. The induced eddy current is generated inside the base material which is located in the alternating magnetic field. The base material can obtain the energy and be heated by the electromagnetic induction [47]. At the initial time of the welding process, the magnetic flux density distribution and current density distribution in the longitudinal section of base material under the electromagnetic parameters of current intensity 3500 A and current frequency 100 kHz during LIHW are shown in Fig. 10. From Fig. 10(a), the magnetic flux density gradually decreases as the distance from the lower surface of base material increases. From Fig. 10(b), the current density decreases exponentially as the distance from the lower surface of base material increases. This is due to that the eddy current generated by electromagnetic induction mainly concentrated on the lower surface of base material with the influence of the skin effect [47].

Compared to the dynamic behaviors of molten pool and keyhole during SLW, that during LIHW are significantly affected by the electromagnetic induction heating. The dynamic behaviors of molten pool and keyhole during LIHW are shown in Fig. 11. At t_1 , the Joule heat

generated by eddy current has less effect on the temperature distribution of base material since the short electromagnetic induction heating time and the variation of flow field of molten pool is small, as shown in Fig. 11 (a). After that, the temperature at bottom of base material gradually increases as the electromagnetic induction heating time increases. The flow velocity of molten metal at the bottom of molten pool decreases compared to that during SLW, as shown in Fig. 9(b) and Fig. 11(b). However, the temperature of molten pool varies slightly because the current density in upper part of base material is greatly less than that in lower part of base material, as shown in Fig. 10(b). At t_1+50 ms, the electromagnetic induction heating indicates the significant effects on the dynamic behaviors of molten pool and keyhole. From Fig. 9(c) and Fig. 11(c), it can be seen that the stability of keyhole during LIHW has been improved compared to that during SLW. The obvious collapse is not formed at bottom of keyhole during LIHW. At t_1+80 ms, the uniformity of temperature distribution in the base material has been improved and the length of the rear part of molten pool increases continuously, as shown in Fig. 11(d). During the LIHW process, the keyhole stability has been significantly improved, which can prevent the formation of pores in weld. This is because that electromagnetic induction heating inhibits the collapse at bottom of keyhole effectively.

The variation curves of molten pool depth and half width with time during SLW and LIHW are shown in Fig. 12. From Fig. 12(a), the molten pool depths during SLW and LIHW increase rapidly firstly and then fluctuate within a certain range, which results from that the shapes of keyhole keep fluctuating during SLW and LIHW. Within 50 ms of the

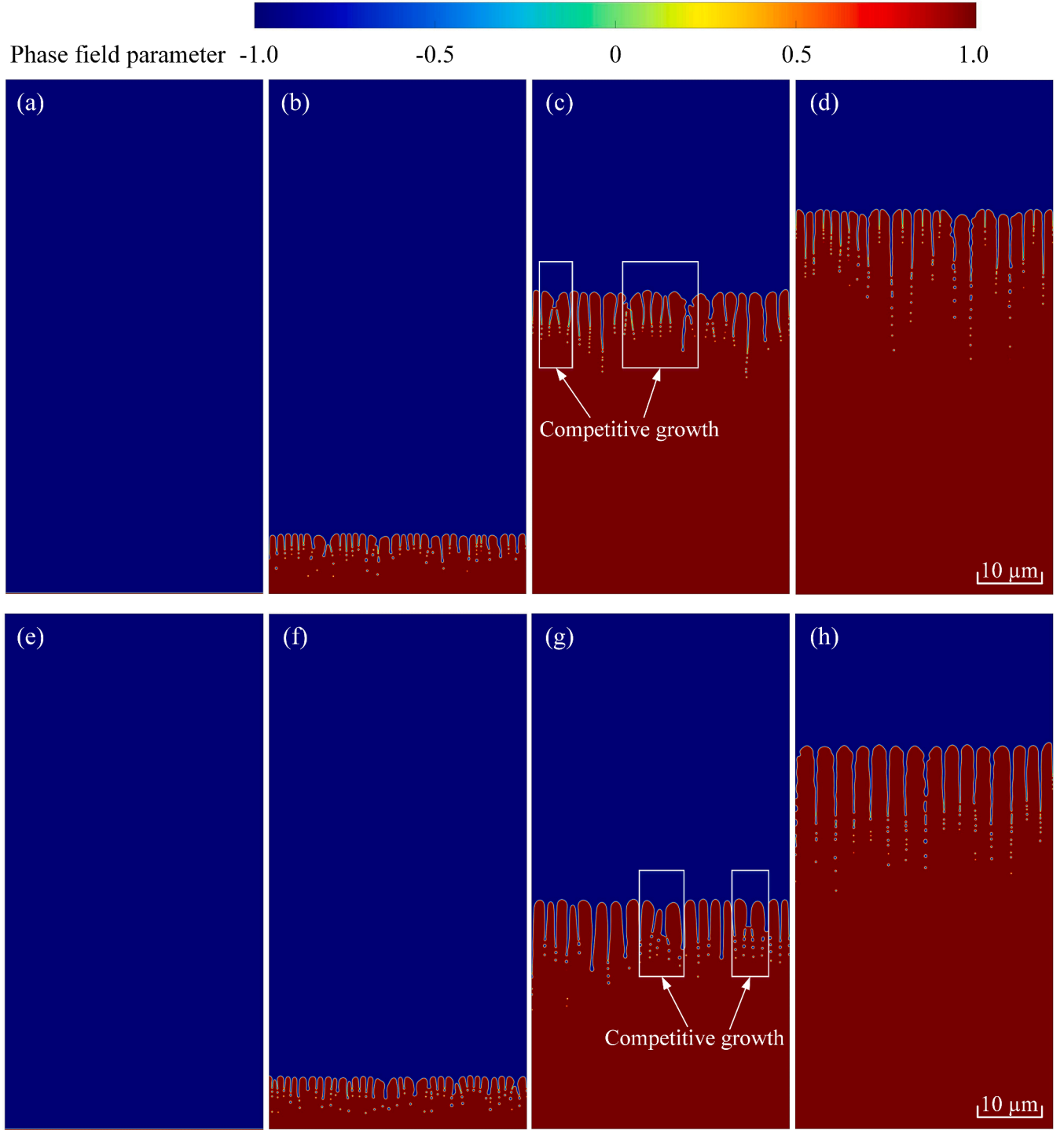


Fig. 15. The microstructure evolution processes during SLW and LIHW: (a, e) t_0 ; (b, f) $t_0 + 0.5$ ms; (c, g) $t_0 + 4.3$ ms; (d, h) $t_0 + 12.8$ ms.

start of welding process, the influence of electromagnetic induction heating on the molten pool depth is weak. The molten pool depths and their varying trends during SLW and LIHW are very similar. After 50 ms, the molten pool depth during LIHW is becoming larger than that during SLW. This is due to the fact that the temperature in lower part of base material increases under the condition of electromagnetic induction heating. From Fig. 12(b), the molten pool half width during SLW increases rapidly initially and then gradually decreases. After that, the molten pool half width during the SLW process remains in the relatively stable status. Within 50 ms of the start of welding process, the influence of electromagnetic induction heating on the molten pool half width can

be ignored and hence the molten pool half widths during SLW and LIHW are almost the same. After 50 ms, the molten pool half width during LIHW is larger than that during SLW. This is because the more energy from the electromagnetic induction heating is absorbed by the base material and temperature of upper part gradually increases.

3.3. Comparison of transient SCs

Based on the simulation results of temperature field, the transient SCs at the point P_i (Fig. 3) near the fusion line during SLW and LIHW are obtained by the transient SCs model, as shown in Fig. 13. During SLW

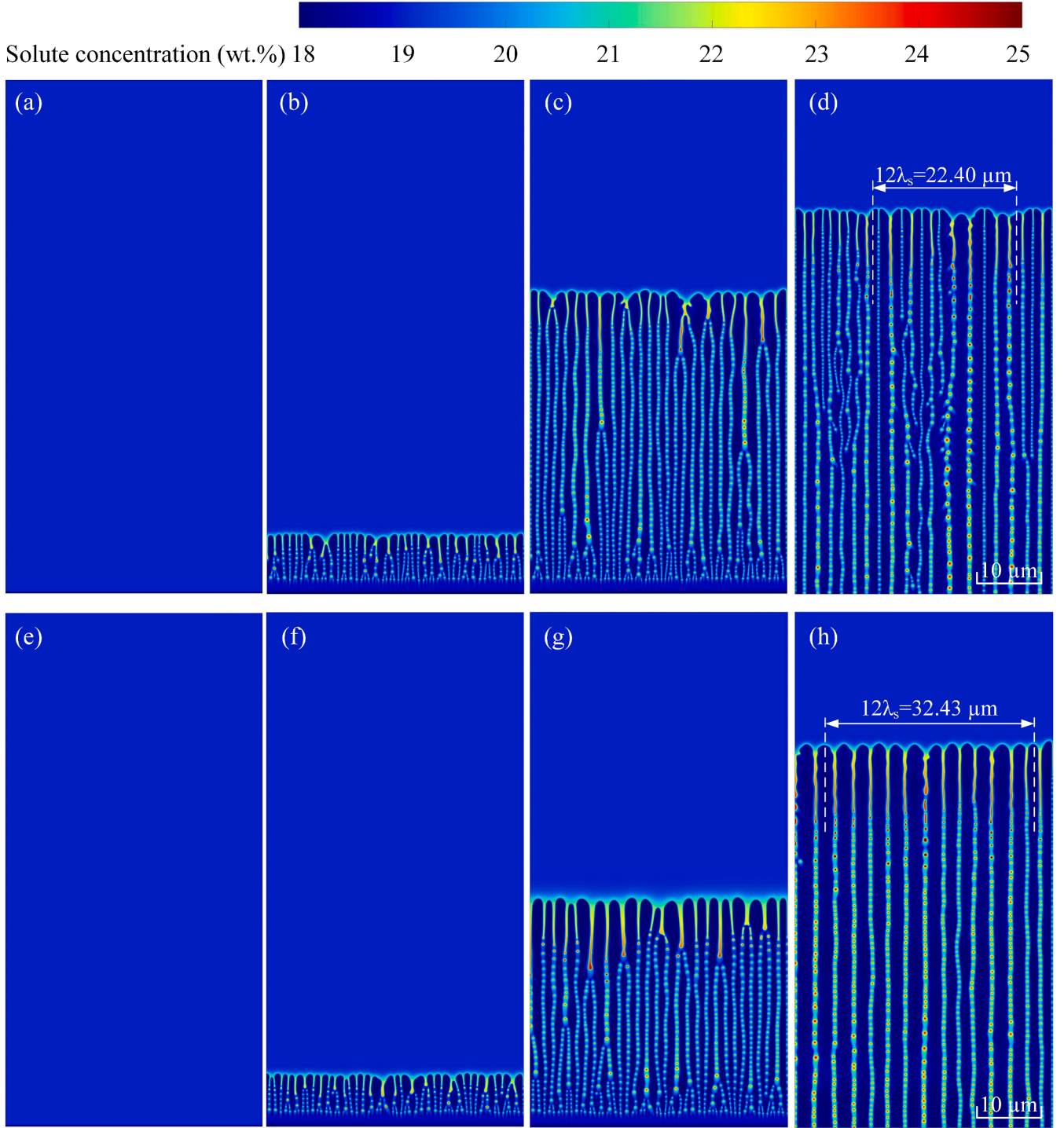


Fig. 16. The evolution processes of solute field during SLW and LIHW: (a, e) t_0 ; (b, f) $t_0+0.5$ ms; (c, g) $t_0+4.3$ ms; (d, h) $t_0+12.8$ ms.

and LIHW, the point P_i gradually moves to the rear part of the molten pool along the fusion line. The transient TG and SR at the point P_i near the fusion line during the solidification process of molten pool are extracted. From Fig. 13(a), the TGs near the fusion line during SLW and LIHW are decreasing gradually with fluctuation. This is because the shapes of the molten pool and keyhole keep fluctuating during the laser beam moving process and the position of point P_i relative to the fusion line is changing correspondingly. The average TG during LIHW is larger than that during SLW with the effect of electromagnetic induction heating increasing. From Fig. 13(b), the SRs near the fusion line during SLW and LIHW gradually increase with fluctuation. According to Eq.

(24), the variation of SR mainly depends on the angle between TG and welding speed. The SR near the fusion line fluctuates under the effect of the variation of transient TG. And the average SR near the fusion line during LIHW obviously decreases under the effect of electromagnetic induction heating compared to that during SLW, which is because the electromagnetic induction heating results in the length of rear part of molten pool increasing.

The variation curves of the product of TG and SR $G \times R$ with solidification time during SLW and LIHW are shown in Fig. 14(a). From Fig. 14(a), the $G \times R$ decreases first and then increases with fluctuation. The average $G \times R$ during LIHW is lower than that during SLW under the

effect of electromagnetic induction heating. The transient TG and SR during the solidification process of the molten pool have significant effects on the grain morphology and size. While the $G \times R$ mainly affects the grain size and the ratio of TG and SR G/R mainly affects the grain morphology, as shown in Fig. 14(b) [48]. From Fig. 14(b), it can be seen that lower $G \times R$ will lead to an increase in grain size, which will effectively reduce the hardness and enhance the plasticity of welded joints.

3.4. The microstructure evolution process

The microstructure evolution process during SLW obtained by solving the microscopic PFM is shown from Fig. 15(a) to (d). The microstructure evolution process during LIHW obtained by solving the microscopic PFM is shown in Fig. 15(e) to (h). From Fig. 15(a) and (e), the microstructure grows in the form of planar grain and mainly distributes in the initial interface of the molten pool solidification process. As the solidification process progresses, the stability of initial interface gradually decreases and a large number of bulges are formed, as shown in Fig. 15(b) and (f). And the microstructure during SLW grows faster than that during LIHW at $t_0 + 0.5$ ms, which is because the SR during SLW is greater than that during LIHW at the same moment, as shown in Fig. 13(b). From Fig. 15(c) and (g), the microstructure gradually evolves from the structure of bulge and grows in the form of columnar grain. The competitive growth among columnar grains with different sizes appears in this stage. The columnar grains with faster growth rate have larger growth space and the growth of other neighboring columnar grains is inhibited, which results in the difference of columnar grain size. The tip positions of columnar grains during LIHW are behind that during SLW at $t_0 + 4.3$ ms, which is because the SR during LIHW is lower than that during SLW at the same moment, as shown in Fig. 13(b). From Fig. 15(d) and (h), the columnar grains continue to grow forward in the relatively stable state and the tip positions of columnar grains are basically consistent in this stage.

The evolution process of solute field during SLW obtained by solving the microscopic PFM is shown in Fig. 16(a) to (d). The evolution process of solute field during LIHW obtained by solving the microscopic PFM is shown in Fig. 16(e) to (h). From Fig. 16(a), (b), (e) and (f), the solute gradually precipitates from the solid phase region, and the solute enrichment layer is formed on the liquid side of solid-liquid interface. From Fig. 16(c) and (g), a large number of solute-enriched droplets accumulate in the gap of columnar grain. This is because the narrowed intercolumnar liquid phase regions are formed during the columnar grain growth process and the solute is hard to diffuse into the liquid phase region away from columnar grains. From Fig. 16(d) and (h), the primary dendrite arm spacings during SLW and LIHW at $t_0 + 12.8$ ms are about $1.87 \mu\text{m}$ and $2.70 \mu\text{m}$, respectively. This is because the $G \times R$ during SLW is larger than that during LIHW at the same moment under the effect of electromagnetic induction heating, as shown in Fig. 14.

4. Conclusions

A macro-micro numerical model for investigating the effects of electromagnetic induction heating on the molten pool and keyhole dynamic behaviors and weld microstructure during LIHW of stainless steel is developed in this paper. The validity of the developed model is verified by comparing the simulation and experimental results. The results show that the depth and half width of the molten pool are increased and the stability of the keyhole is improved during LIHW compared to that during SLW under the effect of electromagnetic induction heating. As the solidification time goes, the transient TGs gradually decrease with fluctuation and the transient SRs gradually increase with fluctuation during SLW and LIHW. The average TG during LIHW is larger than that during SLW and the average SR during LIHW is lower than that during SLW under the effect of electromagnetic induction heating. The primary dendrite arm spacing during LIHW is larger than that during SLW due to the decrease of $G \times R$.

CRediT authorship contribution statement

Yuewei Ai: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Yang Zhang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Shibo Han:** Writing – review & editing, Validation, Formal analysis. **Xin Liu:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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